MALEK BENCHEHIDA

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TECHNICAL SKILLS

- **Languages:** Python, C++, Rust, Go, TypeScript, JavaScript, C, Java, SQL
- **Frameworks & Libraries:** Next.js 15, React 19, Node.js, Drizzle ORM, Zod, Tailwind CSS, LightGBM
- **Tools & Hardware:** AWS, Supabase, Linux, Git, GitHub Actions, I2C, CAN Bus, NVIDIA Jetson AGX Thor

WORK EXPERIENCE

MatchNice Chicago, IL (Remote)
Software Engineer **June 2026 – Present**

- Build a donation platform using Next.js 15, React 19, and TypeScript in a pnpm monorepo.
- Integrate Stripe, Givebutter, and FundraiseUp webhooks into Supabase via Drizzle ORM.
- Automate API code generation using OpenAPI and Orval for type-safe React Query hooks.
- Streamline feature development using AI-assisted pipelines in Replit.

CAS Enterprise Inc. Chicago, IL
DevOps & Full Stack Developer **Aug 2023 – June 2026**

- Managed AWS/Azure cloud infrastructure to ensure high availability for production applications.
- Automated CI/CD pipelines using Python to reduce deployment latency and increase shipping speed.
- Monitored performance with custom scripts to eliminate bottlenecks and backend downtime.

University of Illinois Chicago (UIC) Chicago, IL
Undergraduate Software Research Intern **Jun 2024 – Jul 2024**

- Evaluated security vulnerabilities and prompt injection risks in LLM implementations.
- Wrote Python validation scripts to evaluate model robustness and protect consumer PII.

PROJECT EXPERIENCE

EdgeGuardNano Intrusion Detection System **2026**
Embedded Machine Learning Developer

- Engineered a real-time, offline intrusion detection system deployed directly onto an NVIDIA Jetson AGX Thor edge device.
- Optimized LightGBM machine learning models for ARM64 CPU architecture, achieving a 632x throughput advantage over traditional cloud-based equivalents.
- Designed asynchronous data pipelines for high-throughput sensor monitoring and deterministic, low-latency execution.

SAE Formula Electric Team — UIC **2024 – Present**
Embedded Systems Software Engineer

- Programmed real-time embedded control loops in C++ for motor speed and battery optimization.
- Migrated performance-critical firmware from C++ to Rust for memory safety and race-free concurrency.
- Developed communication interfaces between ECUs and sensors over the CAN Bus protocol.

Personal Portfolio & Cyber Briefing Dashboard **2026**
Full Stack Developer

- Developed and deployed a responsive personal portfolio website hosted via GitHub Pages.
- Integrated a live Cyber Briefing dashboard aggregating cybersecurity news with automated daily content refreshes using continuous integration pipelines via GitHub Actions.

Reaction-Time Arena **2026**
Hardware & Embedded Systems Developer

- Built a multiplayer hardware system utilizing three microcontrollers communicating via the I2C protocol.
- Designed a master-slave architecture to process low-latency inputs and execute real-time scoring logic.

EDUCATION

University of Illinois Chicago Chicago, IL
Bachelor of Science in Computer Science **Expected Graduation: December 2026**

- **Activities:** Technical Software Lead, SAE Formula Electric Team